

**Genome sequencing of ‘Fuji’ apple clonal varieties reveals genetic mechanism
of the spur-type morphology**

Cai et al.

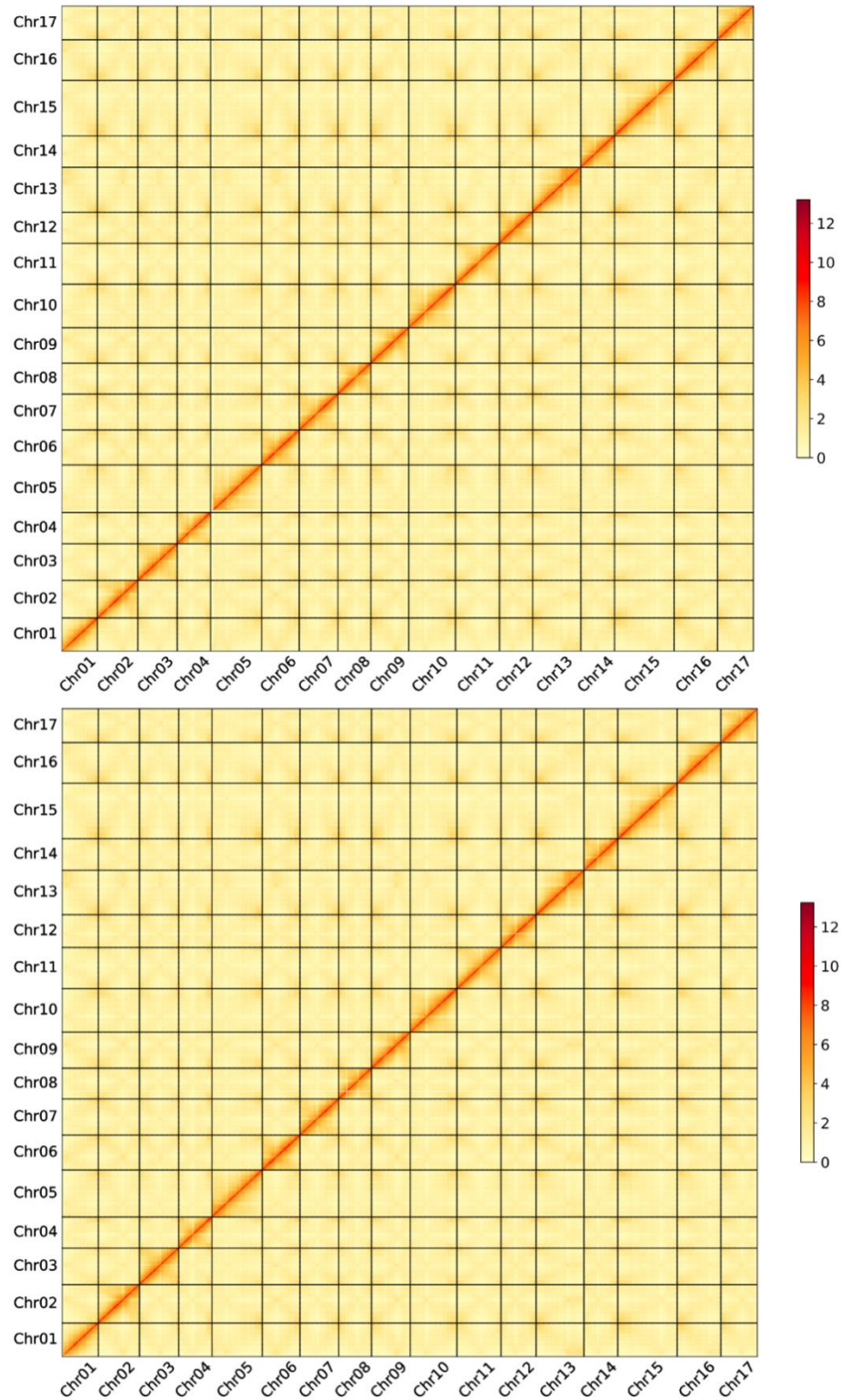
Supplementary Note 1. K-mer-based quality assessment of the ‘Fuji’ genome assembly

To evaluate the quality of the haplotype-resolved assembly of ‘Fuji’, we conducted a systematic k-mer-based quality assessment. We also included two recently published ‘Fuji’ phased assemblies, generated using the HiFi+Hi-C strategy, for comparison (hereafter referred to as FujiAsm1¹ and FujiAsm2²). The k-mers used for the analysis were generated from the raw sequencing data of ‘Delicious’, ‘Ralls Janet’, and ‘Fuji’. First, we analyzed the distribution of k-mers generated from the Fuji sequencing data in the haplotype-resolved assemblies (Supplementary Fig. 4a-c). In all three assemblies, read-only k-mers were present at low frequencies, which typically represented sequencing errors in the reads, consistent with the characteristics of a correctly assembled genome. The three assemblies exhibited a single 1-copy peak, representing haplotype-specific sequences that were expected to be assembled only once, indicating that these haplotype-specific sequences were correctly assembled. When further analyzing the distribution of these haplotype-specific k-mers in the two haploid genomes of the assemblies, we observed that half of the k-mers were present only in one haploid genome, while the other half were present only in the other haploid genome (Supplementary Fig. 4d-f). This indicates that these haplotype-specific sequences were evenly distributed between the two haploid genomes of each assembly.

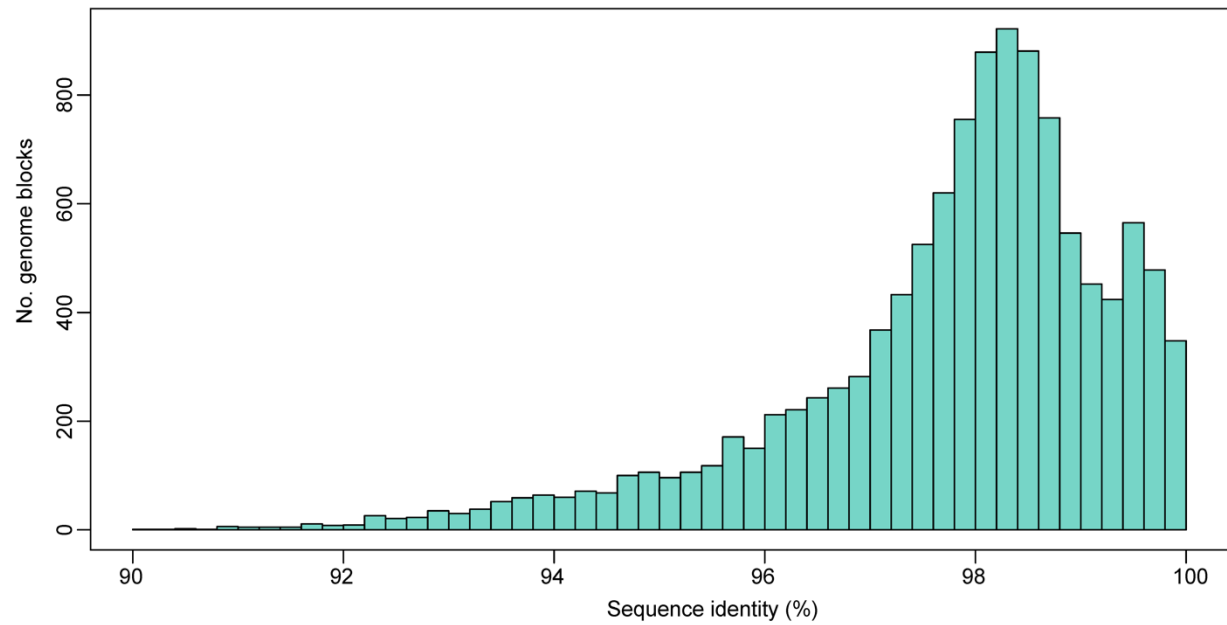
In our assembly and FujiAsm1, the 2-copy peak exhibited a single peak, indicating that these 2-copy sequences were correctly assembled. However, in FujiAsm2, the 2-copy peak also included small 1-copy and 3-copy peaks, indicating that a small portion of 2-copy sequences in the genome were assembled as 1-copy and 3-copy sequences (Supplementary Fig. 4a-c). When further examining the distribution of k-mers in the 2-copy peak across the two haploid genomes (Supplementary Fig. 4d-f), we observed that most 2-copy k-mers were present in both haploid genomes, meaning each haploid genome contained one copy. However, we also observed small haplotype-specific k-mer peaks within the 2-copy peak, indicating the presence haplotype-specific duplications, which were present in two copies on the same haplotypes. Overall, the k-mer spectrum of our assembly aligns with the characteristics of a correctly assembled haplotype-resolved genome.

We further evaluated genome phasing accuracy by analyzing the distribution of ‘Ralls Janet’ and ‘Delicious’ haplotype-specific k-mers (hap-mers) in the phased assemblies. Specifically, ‘Delicious’ hap-mers refer to k-mers present in both the ‘Delicious’ and ‘Fuji’ sequencing reads but absent in the ‘Ralls Janet’ sequencing reads, and vice versa. In our phased assembly, ‘Ralls Janet’ hap-mers were mainly present in Fuji_Ral (a single red peak in Supplementary Fig. 5a) and were almost absent in Fuji_Del (a single black peak in Supplementary Fig. 5b), while ‘Delicious’ hap-mers were mainly present in Fuji_Del but almost absent in Fuji_Ral (Supplementary Fig. 6a,b). However, in both FujiAsm1 and FujiAsm2, ‘Ralls Janet’ and ‘Delicious’ hap-mers were abundantly present in both haploid genomes of each assembly (mixed black and red peaks in Supplementary Fig. 5c-d, Fig. 6c-d), indicating that these two assemblies were not well phased at the genome level. Further analysis of the distribution of these hap-mers across different chromosomes revealed that all three genomes exhibited a high degree of phasing accuracy within chromosomes. However, in FujiAsm1 and FujiAsm2, we found that different chromosomes within a single haploid assembly originated from different parents (Supplementary Fig. 7), further suggesting that these two assemblies were not well phased at the genome level. However, in our assembly, by employing the trio-binning approach, we could confidently assign the assembled chromosomes to the correct haplotypes. Finally, we examined the parental origin of genomic regions in the three assemblies based on hap-mers and found that our assembly contained 30 potential phase-switching errors, affecting a total of 25.95 Mb sequences (1.96% of the assembly). FujiAsm1 and FujiASM2 included 39 and 34 phase-switching errors, respectively, involving a total of 29.62 Mb and 27.97 Mb sequences (Supplementary Fig. 8).

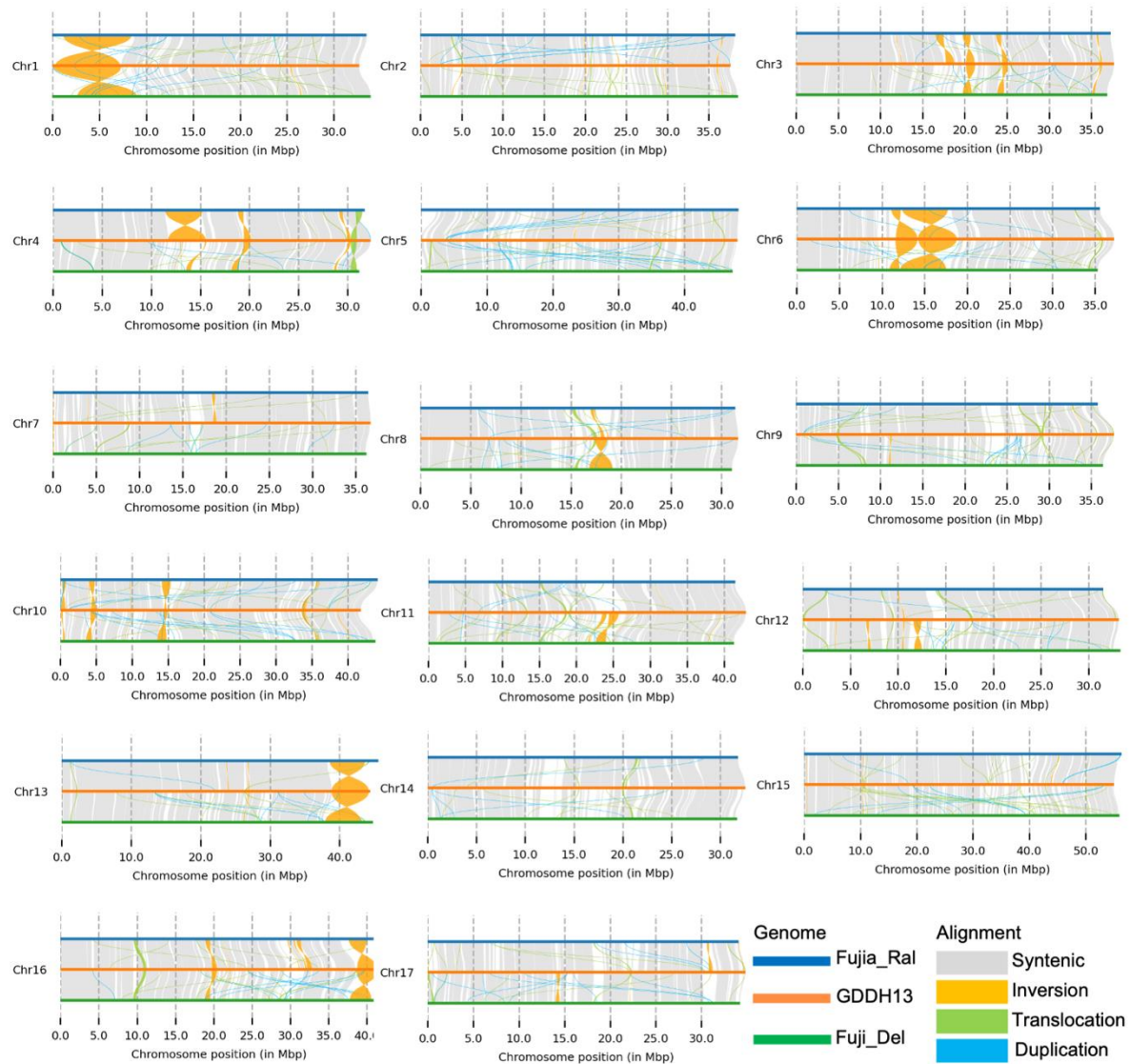
In summary, our comprehensive assessments suggest that all three ‘Fuji’ haplotype-resolved assemblies exhibited similar phasing accuracy at the chromosomal level. The main advantage of the trio-based method used in our assembly is its ability to assign chromosomes of the same origin to the correct haploid genomes.



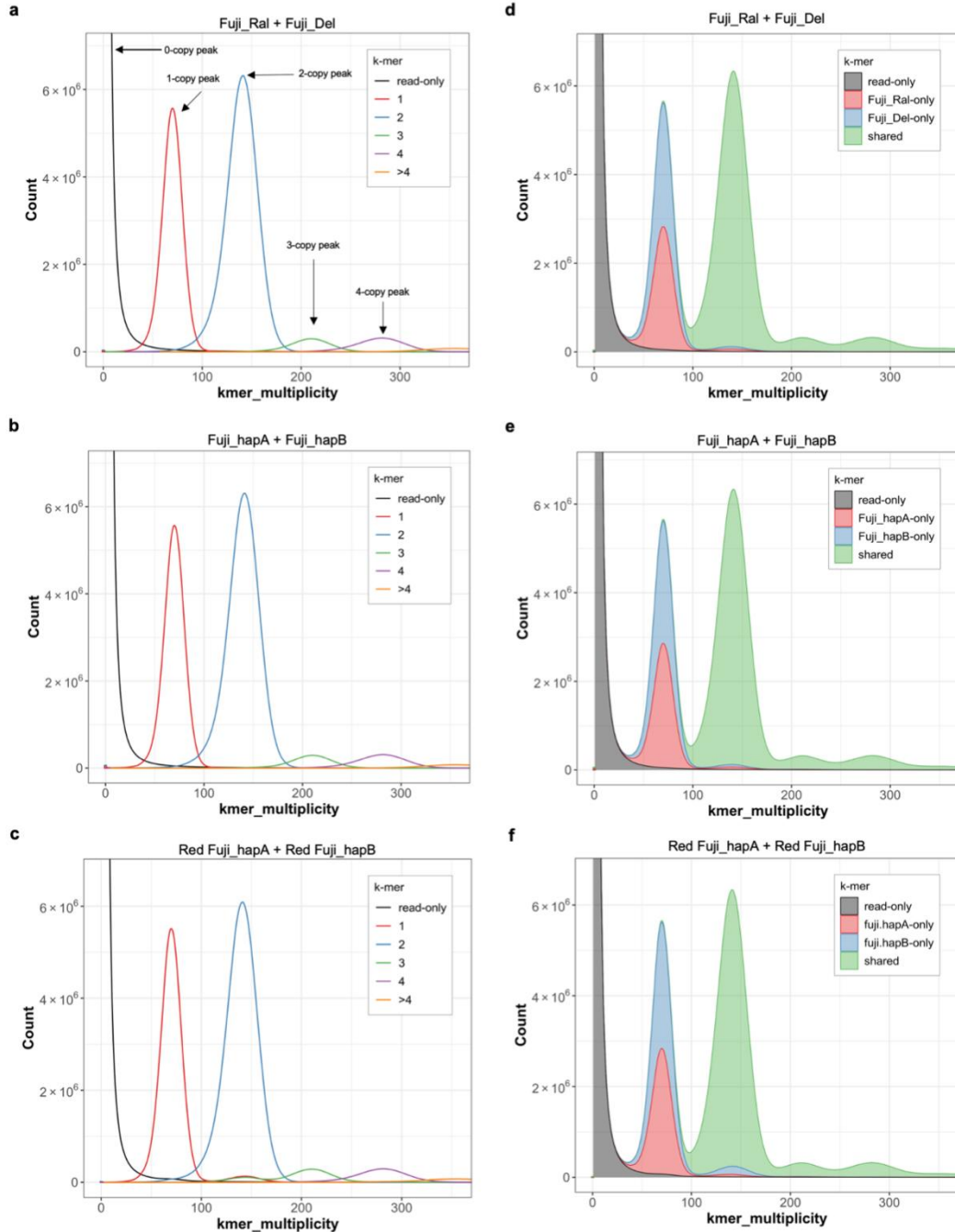
Supplementary Figure 1. Hi-C contact map of Fuji_Ral (top) and Fuji_Del (bottom) assemblies. The intensity of interactions was calculated using a bin size of 600 kb.



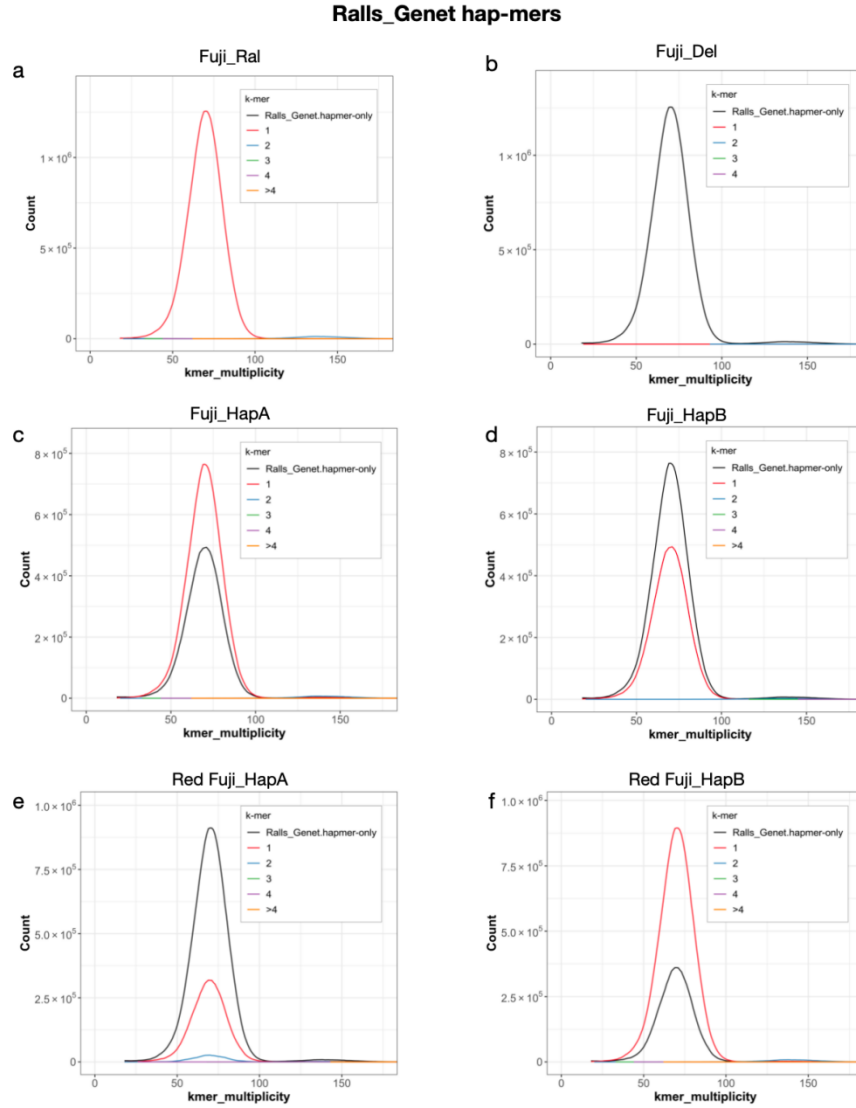
Supplementary Figure 2. Distribution of sequence identities in the alignment blocks between the two haploid genomes of 'Fuji'. The x-axis represents sequence identity, and the y-axis indicates the number of windows with the corresponding sequence identity.



Supplementary Figure 3. Collinearity between the 'Fuji' and GDDH13 genomes.

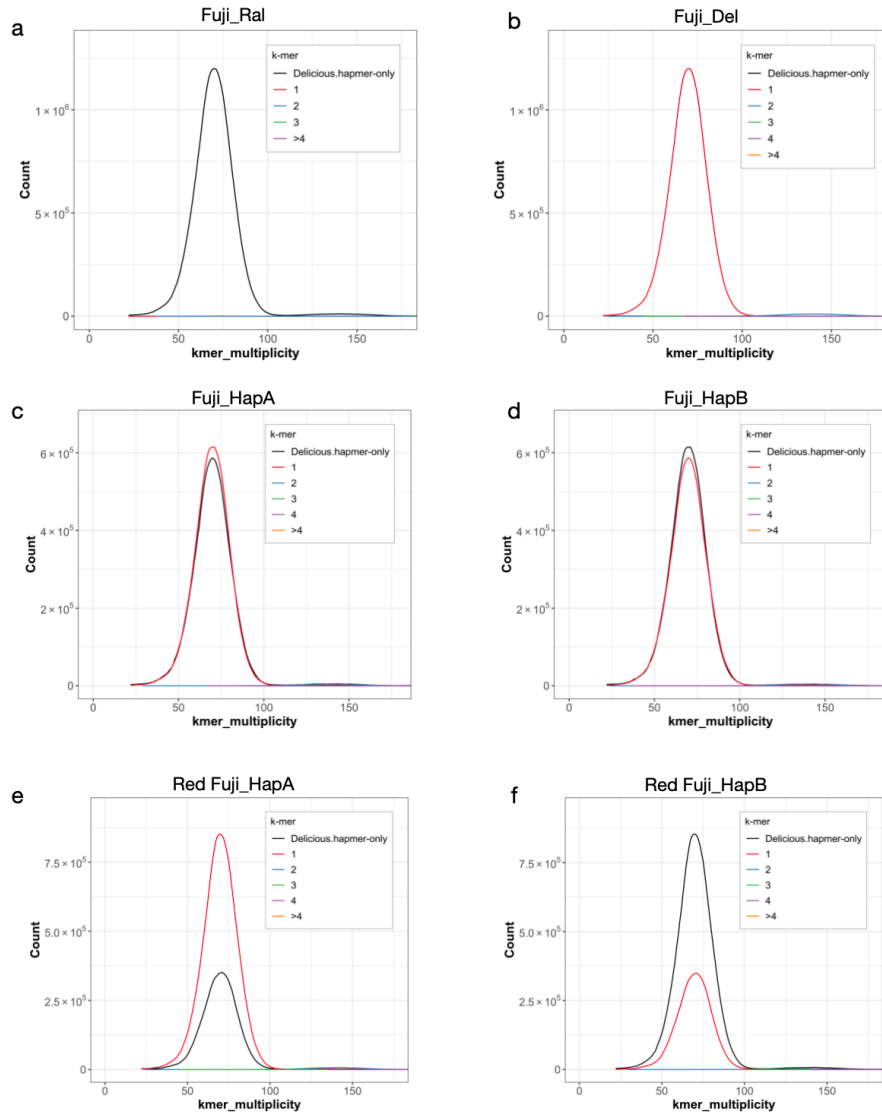


Supplementary Figure 4. K-mer spectrum analysis. **a-c** K-mer comparison plot. The x-axis represents the frequency of k-mers in the HiFi sequencing reads, with different peaks indicating the expected number of occurrences in the genome assembly (0- to 4-copy peaks). Different colors indicate the number of times a k-mer is expected to occur in a given assembly. **d-f** Distinct k-mer spectrum plot of the three phased assemblies of 'Fuji'. The haplome-specific k-mers are colored in red and blue, while the common k-mers are colored in green. Fuji_Ral and Fuji_Del, the two haploid genome assemblies of 'Fuji' generated in this study. Fuji_HapA and Fuji_HapB, the two haploid genome assemblies of 'Fuji' reported in Li *et al.*¹. Red Fuji_HapA and Red Fuji_HapB, the two haploid genome assemblies of 'Fuji' reported in Peng *et al.*².

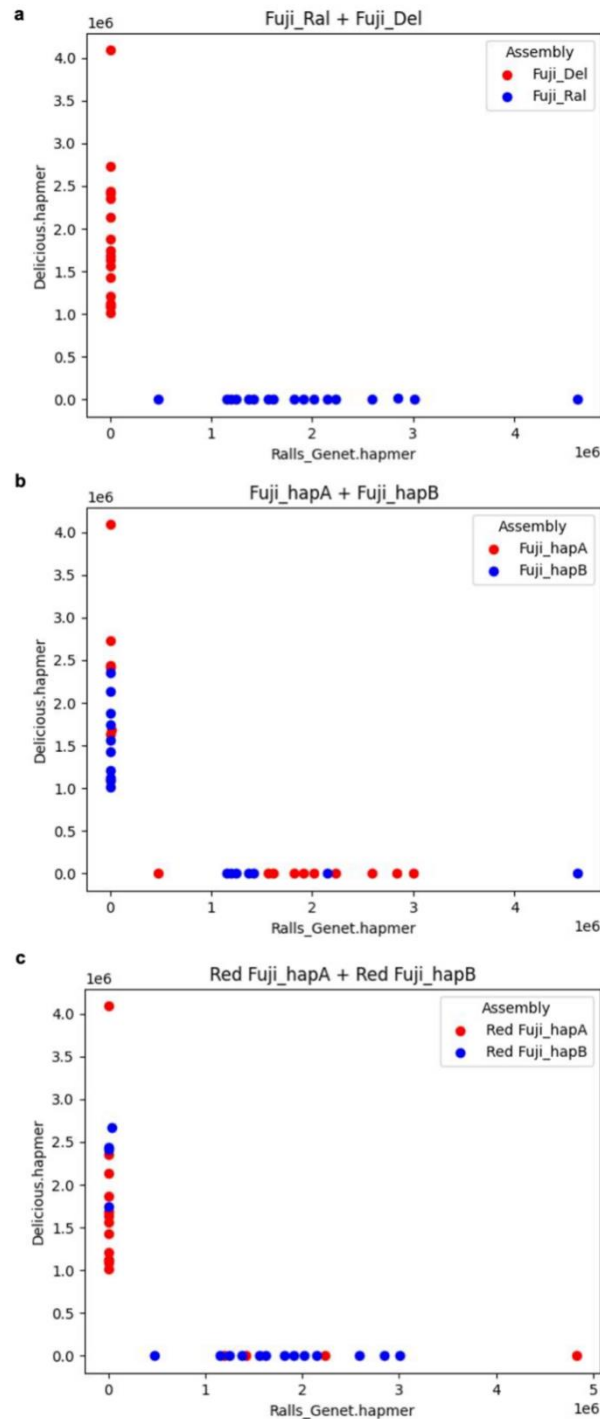


Supplementary Figure 5. Distribution of Ralls Genet hap-mers (Haplotype-specific k-mer) in the three phased ‘Fuji’ genome assemblies. **a,b** Fuji_Ral and Fuji_Del, the two haploid genome assemblies of ‘Fuji’ generated in this study. **c,d** Fuji_HapA and Fuji_HapB, the two haploid genome assemblies of ‘Fuji’ reported in Li *et al.*¹. **e,f** Red Fuji_HapA and Red Fuji_HapB, the two haploid genome assemblies of ‘Fuji’ reported in Peng *et al.*².

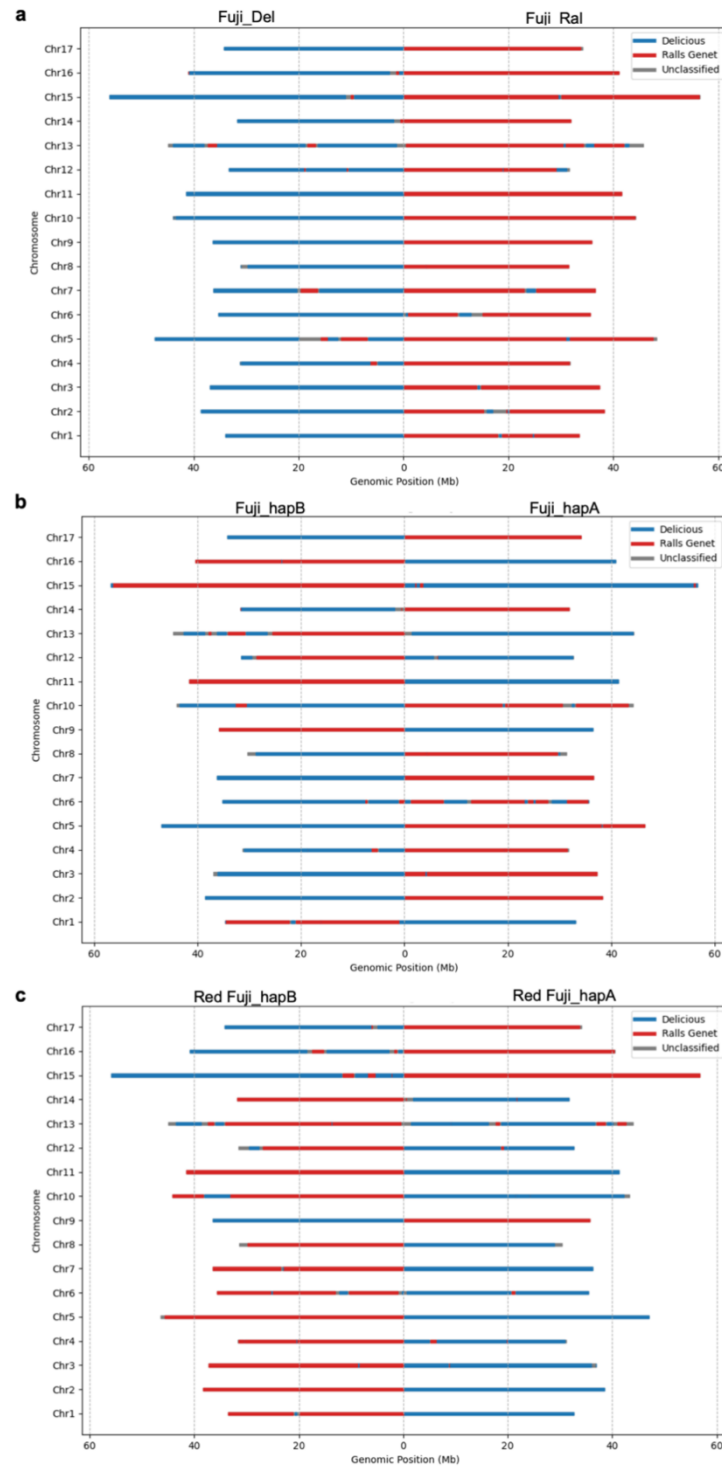
Delicious hap-mers



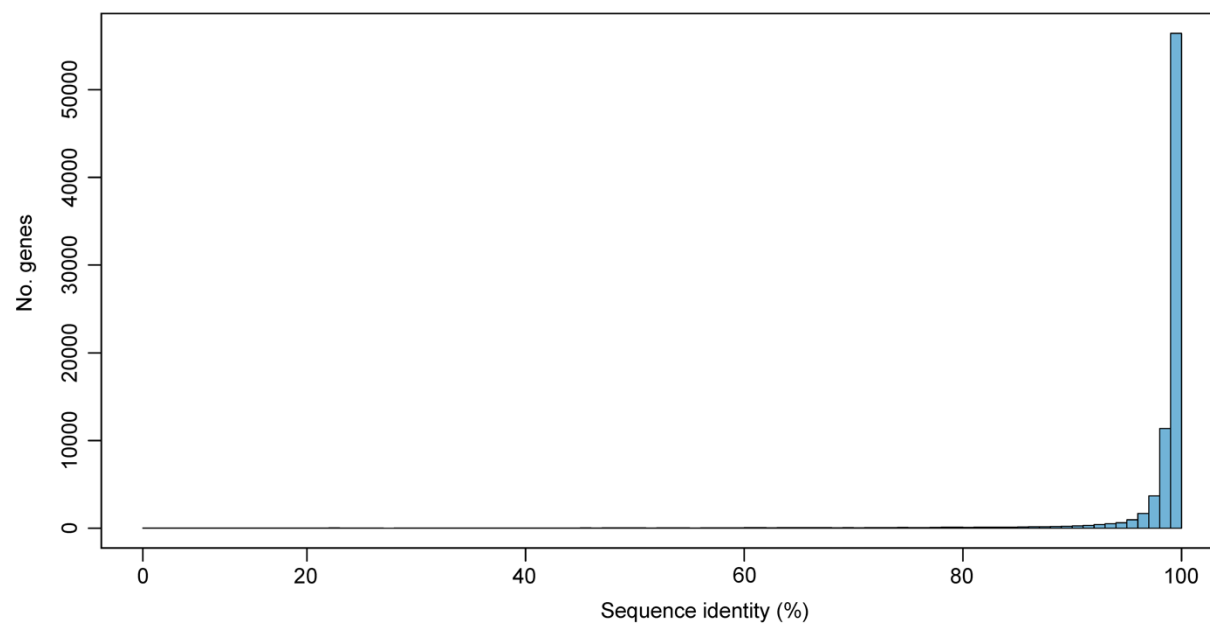
Supplementary Figure 6. Distribution of Delicious hap-mers in the three phased ‘Fuji’ genome assemblies. **a,b** Fuji_Ral and Fuji_Del, the two haploid genome assemblies of ‘Fuji’ generated in this study. **c,d** Fuji_HapA and Fuji_HapB, the two haploid genome assemblies of ‘Fuji’ reported in Li *et al.* ¹. **e,f** Red Fuji_HapA and Red Fuji_HapB, the two haploid genome assemblies of ‘Fuji’ reported in Peng *et al.* ².



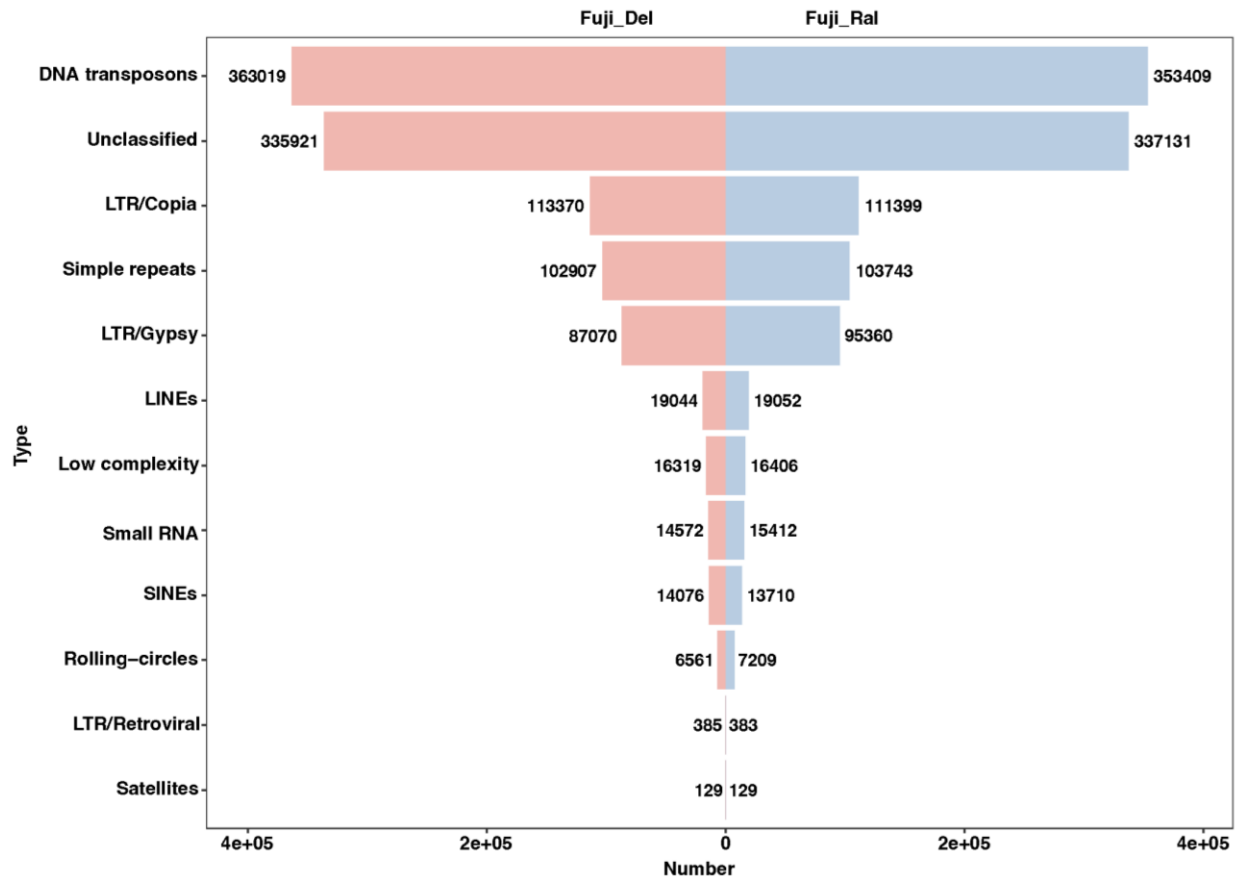
Supplementary Figure 7. Haplotype-specific k-mer (hap-mer) blob plots of ‘Fuji’ phased assemblies. **a-c** Hap-mer blob plots of the assemblies Fuji_Ral and Fuji_Del generated in this study (**a**), Fuji_HapA and Fuji_HapB reported in Li *et al.*¹; (**b**), and Red Fuji_HapA and Red Fuji_HapB, reported in Peng *et al.*²; (**c**). The chromosomes of the two haplomes are colored in red and blue, respectively. The x- and y-axes indicate the number of ‘Ralls Genet’ hap-mers and ‘Delicious’ hap-mers, respectively, present in the chromosomes. Source data are provided as a Source Data file.



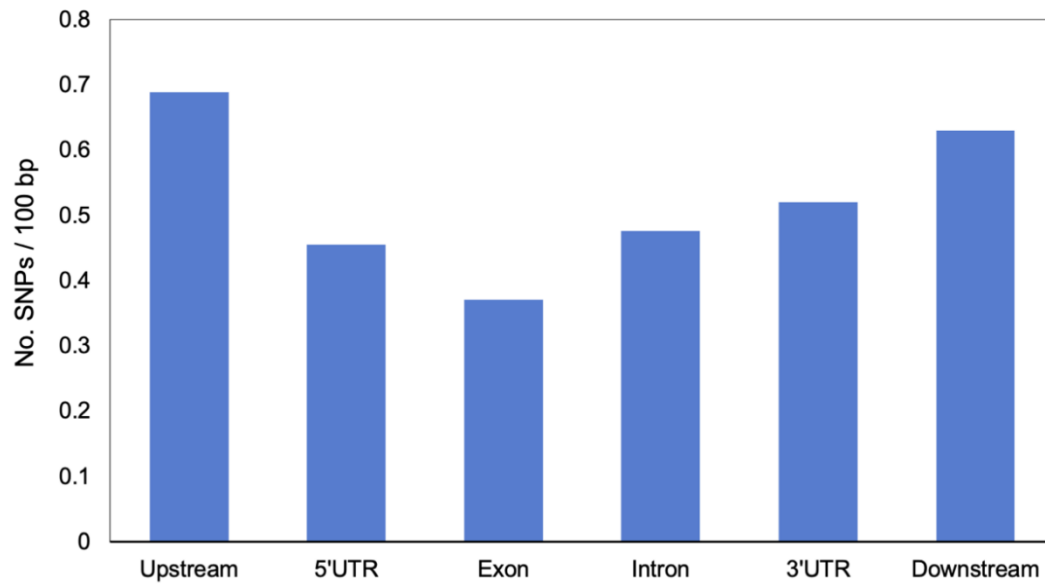
Supplementary Figure 8. Parental origin of genomic regions in ‘Fuji’ phased assemblies. a-c Distribution of genomic regions originating from ‘Ralls Genet’ and ‘Delicious’ in the assemblies Fuji_Ral and Fuji_Del generated in this study (a), Fuji_HapA and Fuji_HapB reported in Li *et al.*¹; (b), and Red Fuji_HapA and Red Fuji_HapB, reported in Peng *et al.*²; (c). Genomic segments originating from ‘Ralls Genet’ and ‘Delicious’, determined based on hap-mers, are marked in red and blue, respectively. Source data are provided as a Source Data file.



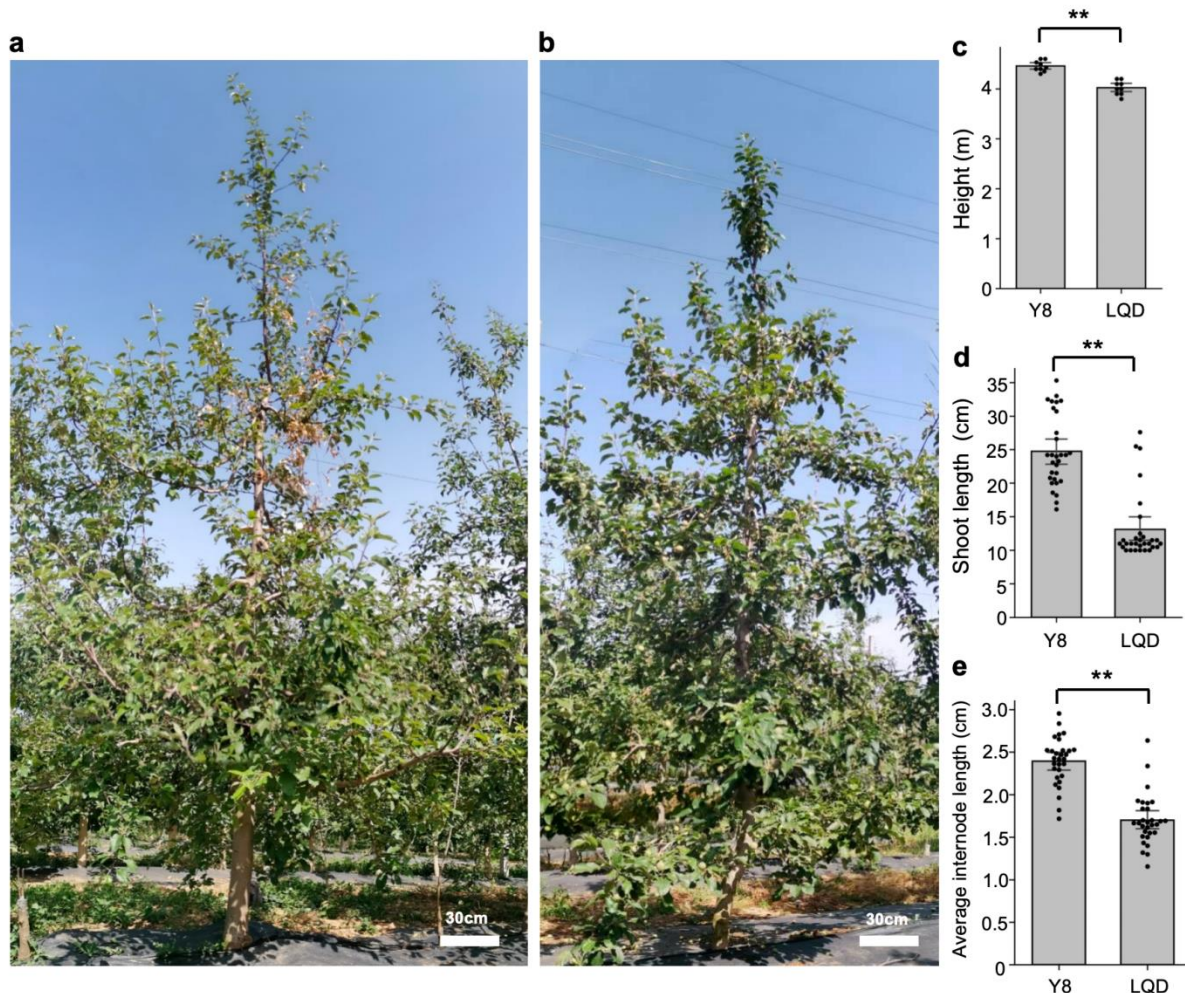
Supplementary Figure 9. Distribution of sequence identities of protein-coding genes between Fuji and GDDH13.



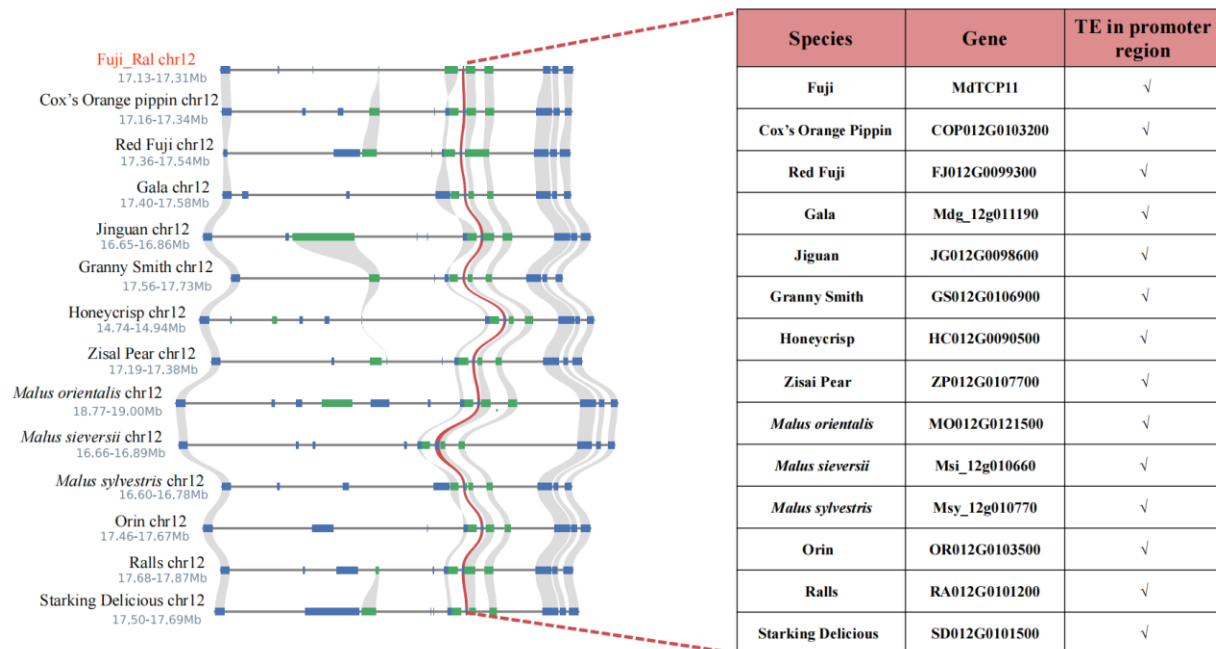
Supplementary Figure 10. Type and number of repeat sequences in the two ‘Fuji’ haploid genomes, Fuji_Del and Fuji_Ral. Source data are provided as a Source Data file.



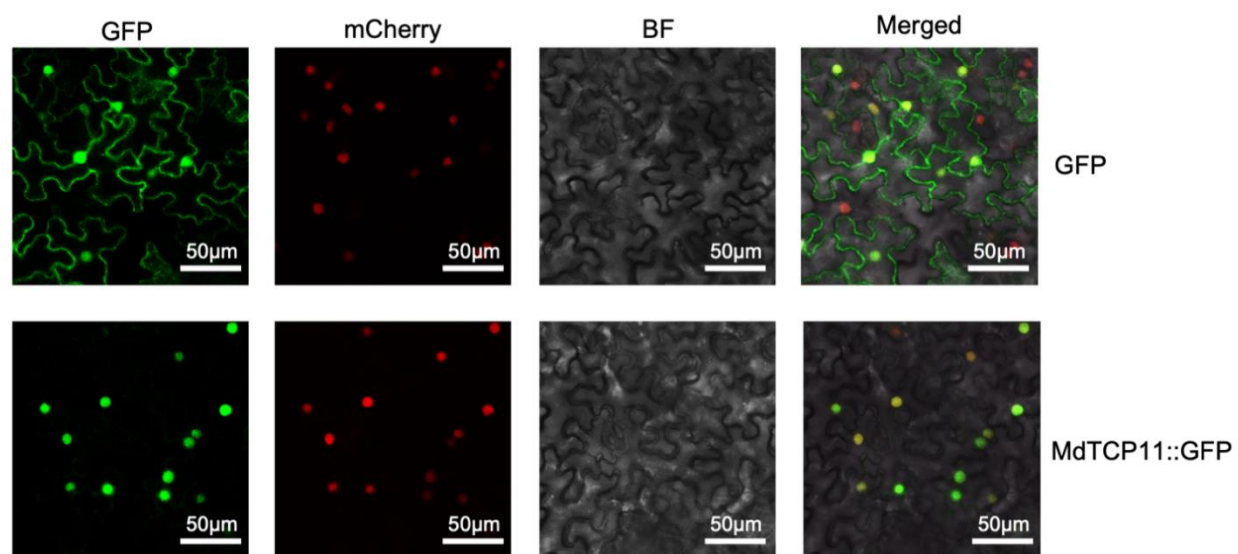
Supplementary Figure 11. Densities of germline SNPs in different gene features. Source data are provided as a Source Data file.



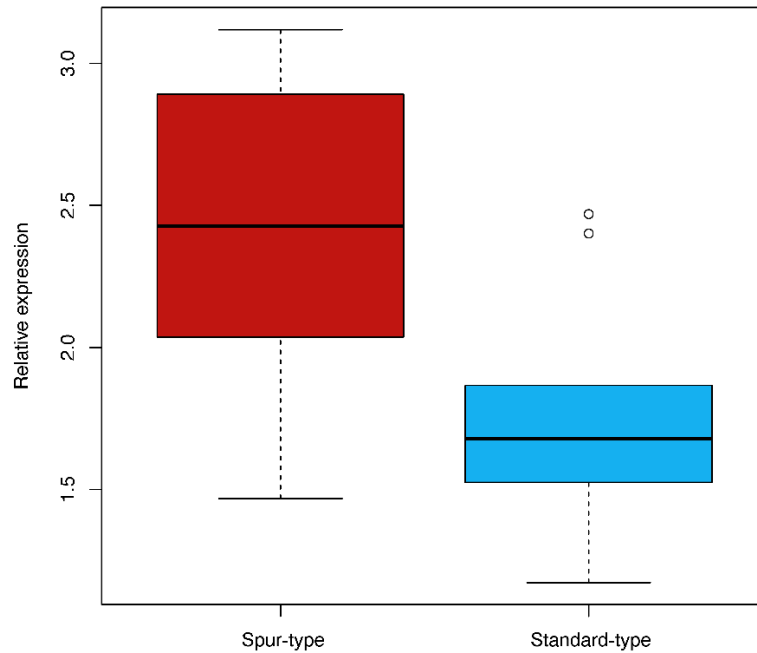
Supplementary Figure 12. Phenotypic characterization of standard-type and spur-type apple varieties. **a,b** Plants of standard-type (**a**) and spur-type (**b**) apple varieties. **c** Height of the standard-type variety ‘Yanfu No.8’ (Y8) and the spur-type variety ‘Liquan spur’ (LDQ). Values are presented as mean $\pm$ SE (n = 9 individual plants). **d** Shoot length LDQ and Y8. Values are presented as mean $\pm$ SE (n = 30 individual annual shoots). Internode length of Y8 and LDQ. Values are presented as mean $\pm$ SE (n = 30 individual annual shoots). In (**b**, **d**, **e**), * and *** indicate significant difference between means ($P < 0.05$ and $P < 0.001$, respectively; two-tailed Student’s *t*-test). Source data are provided as a Source Data file.



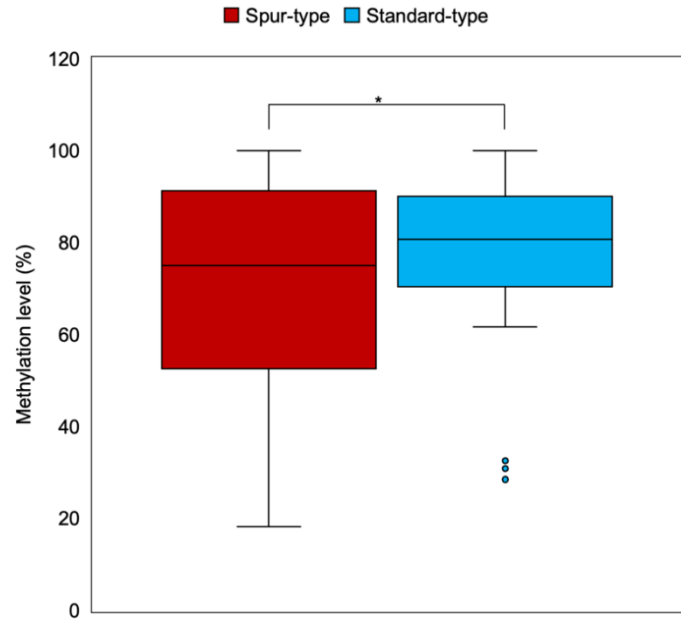
Supplementary Figure 13. Microsynteny of the *MdTCP11* genomic region across ‘Fuji’ genome and 13 other apple accessions. Boxes represent genes, with colors indicating their strand orientation: blue for the forward strand and green for the reverse strand. *MdTCP11* and its homologous genes are connected by red lines, while flanking genes are connected by gray lines. The table on the right shows the presence of the MITE in the promoter of *MdTCP11* across various genomes.



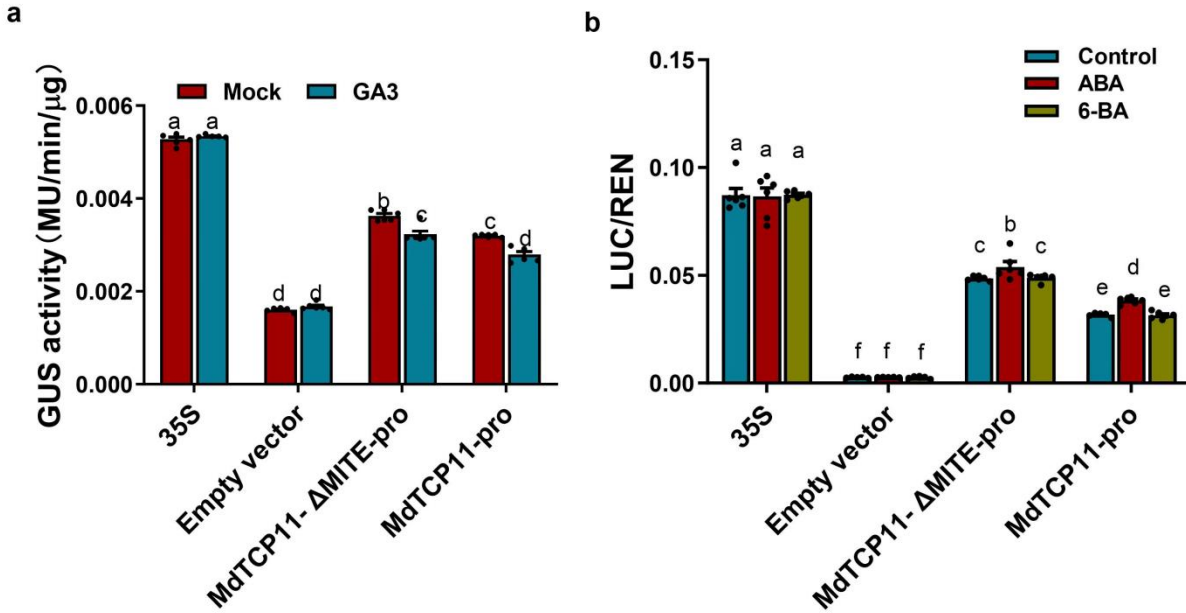
Supplementary Figure 14. Subcellular localization of MdTCP11 in *N. benthamiana* leaves. The nuclei are marked by mCherry fused to a nuclear localization signal. Bars = 50µm.



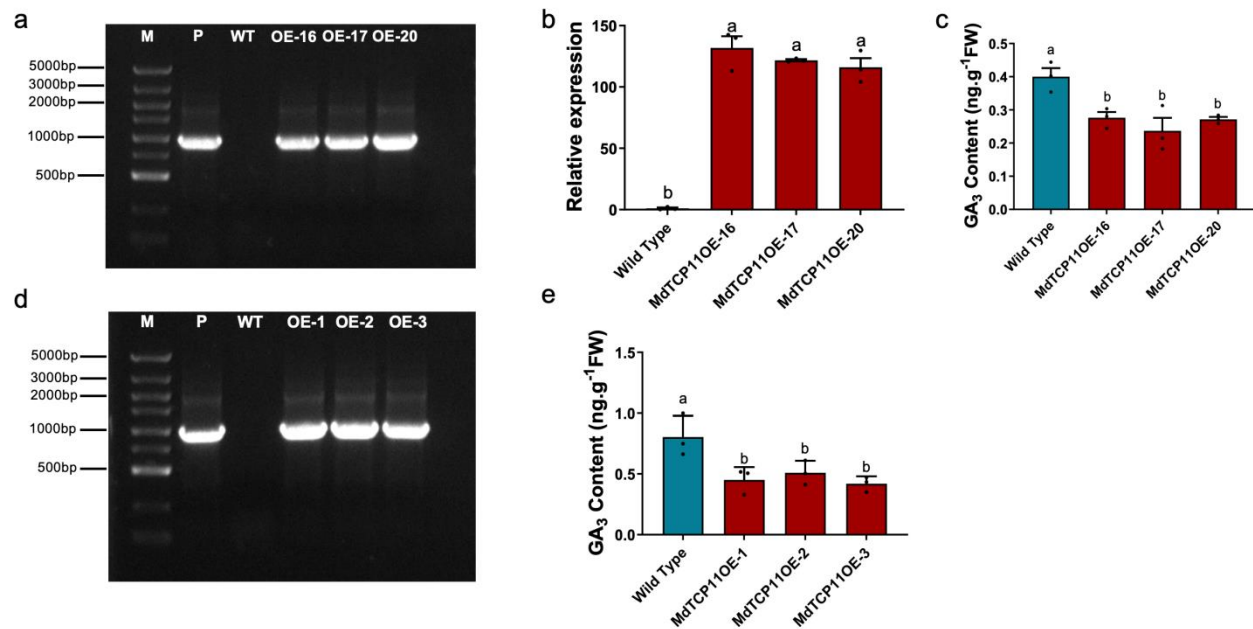
Supplementary Figure 15. Relative expression of *MdTCP11* in shoot tips of nine standard-type and ten spur-type mature apple varieties determined using qRT-PCR. In each box plot, the lower and upper bounds represent the 25th and 75th percentiles, respectively, the center line indicates the median, and the whiskers extend to the full range of the data. Source data are provided as a Source Data file.



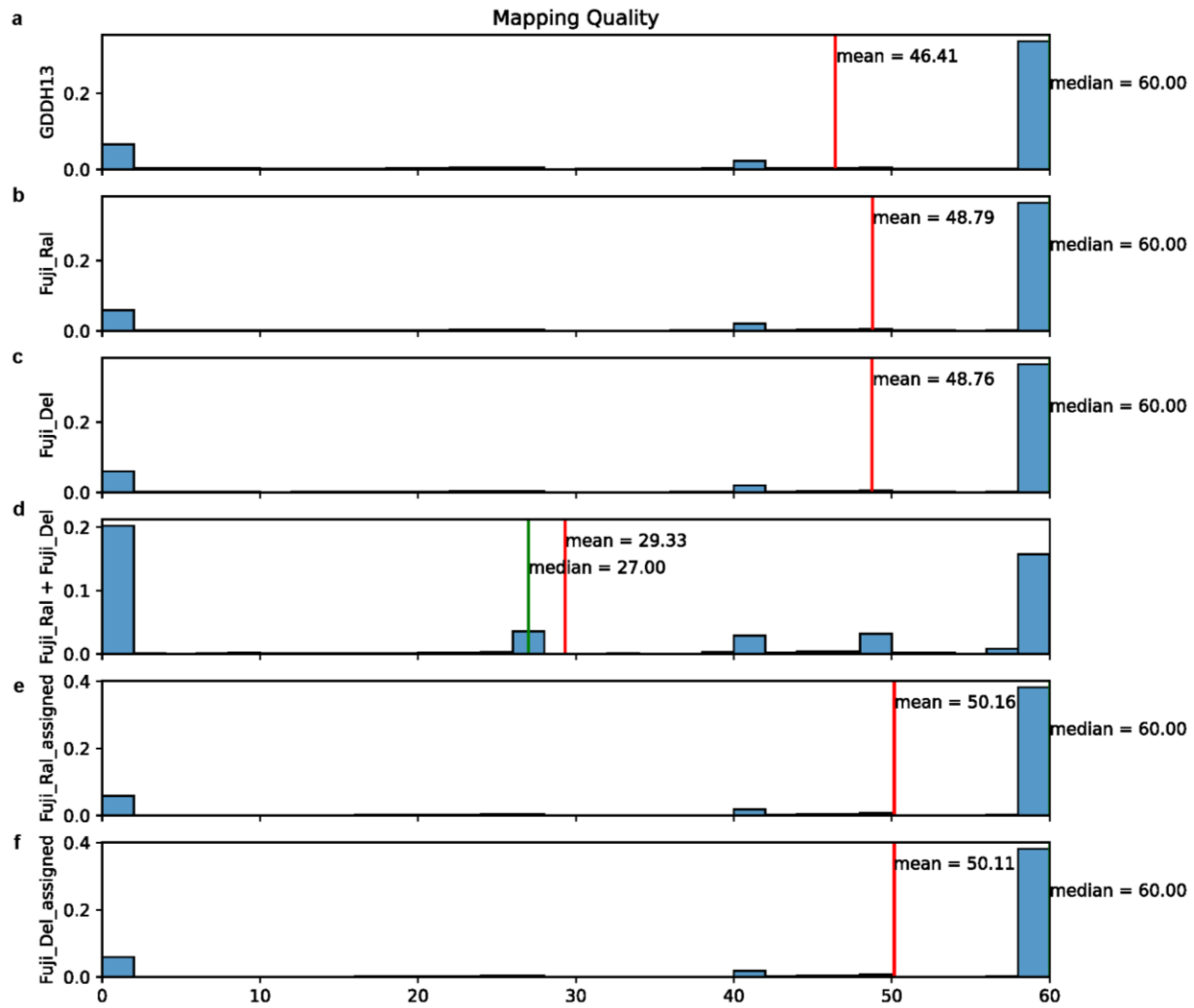
Supplementary Figure 16. Cytosine methylation levels of the MITE in the promoter of *MdTCP11*. Shoot tips from the standard-type ‘Yanfu No.8’ and spur-type ‘Liquan spur’ were analyzed. In each box plot ($n = 32$ sites), the lower and upper bounds represent the 25th and 75th percentiles, respectively, the center line indicates the median, and the whiskers extend to the full range of the data. * denotes a significant difference at $P < 0.05$ (two-tailed Student’s t -test). Source data are provided as a Source Data file.



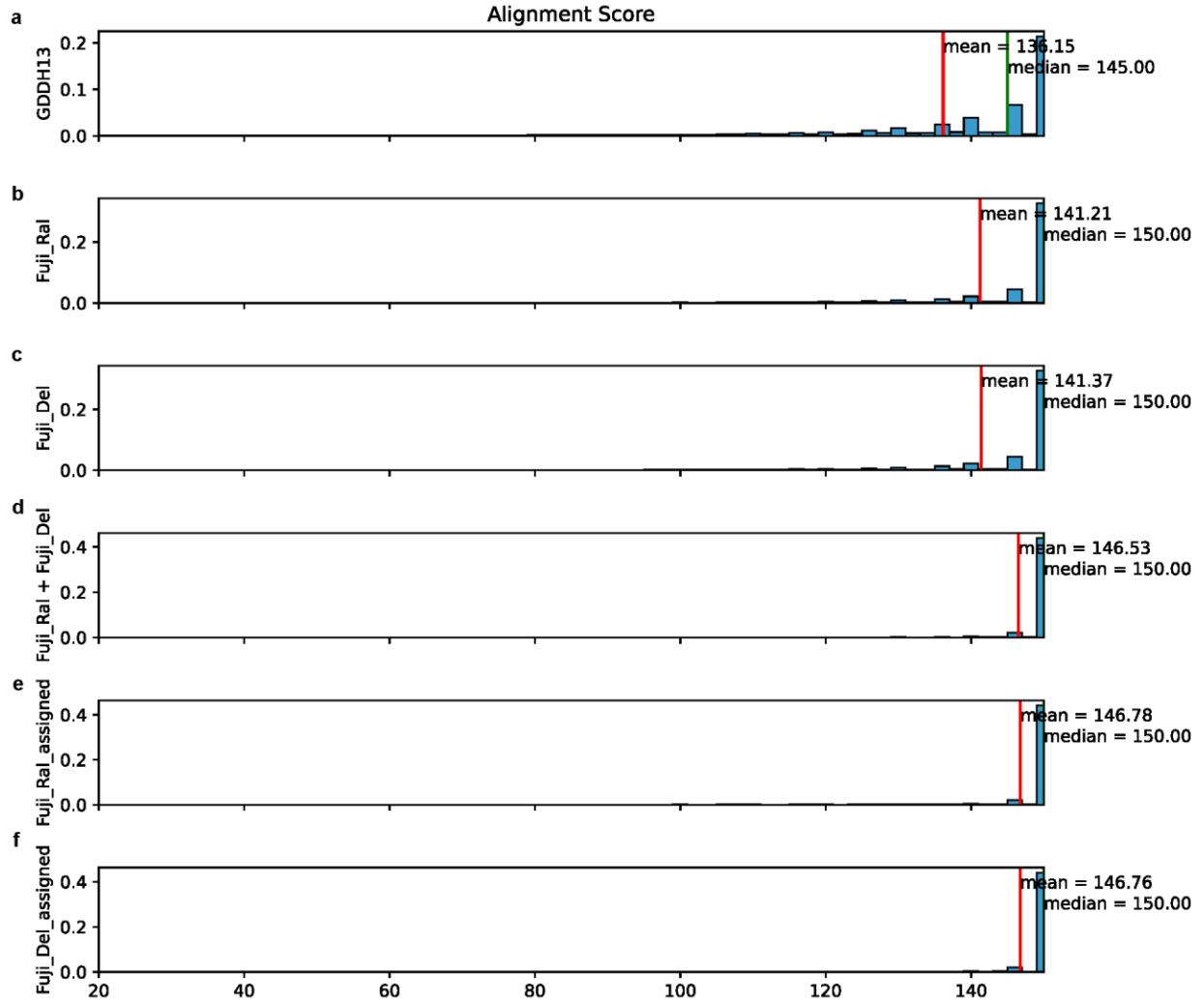
Supplementary Figure 17. Promoter activities of *MdTCP11*. **a** Promoter activity of *MdTCP11* in response to GA3 treatment. TCP11-ΔMITE-pro represents the promoter from the spur-type variety, and TCP11-pro represents the promoter from the standard-type variety. Data are shown as mean ± SE (n = 6 biological replicates). **b** LUC activity of the *MdTCP11* promoter following treatment with cytokinin (6-BA) or ABA. LUC-TCP11-ΔMITE-pro represents the promoter from the spur-type variety, while LUC-TCP11-pro represents the promoter of the standard-type variety. Data are shown as mean ± SE (n = 6 biological replicates). In (**a**, **b**), different letters above the bar graphs indicate significant differences at $P < 0.05$ (One-way ANOVA). Source data are provided as a Source Data file.



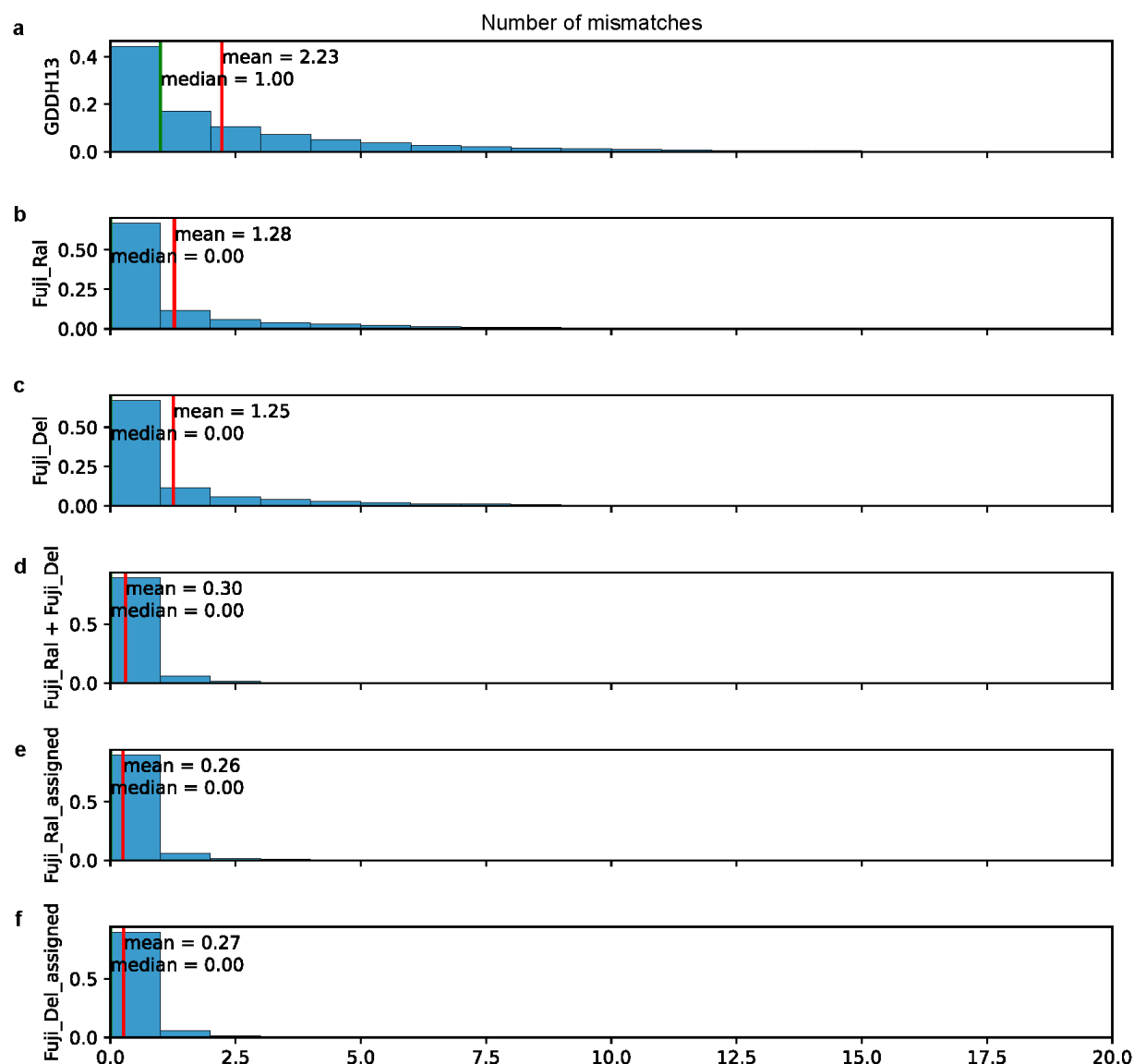
Supplementary Figure 18. Verification of *MdTCP11* overexpression in transgenic apple and tobacco (*Nicotiana tabacum*) lines. **a** Detection of *MdTCP11* genomic DNA in *MdTCP11*-overexpressing tobacco lines. **b** Expression levels of *MdTCP11* in *MdTCP11*-overexpressing tobacco lines. Data are presented as mean $\pm$ SE (n = 3 biological replicates). **c** GA₃ content in transgenic tobacco lines overexpressing *MdTCP11* and wild-type (WT) plants. Data are presented as mean $\pm$ SE (n = 3 biological replicates). **d** Detection of *MdTCP11* genomic DNA in *MdTCP11*-overexpressing apple lines. **e** GA₃ content in transgenic apple lines overexpressing *MdTCP11* and wild-type ‘GL-3’ plants. Data are presented as mean $\pm$ SE (n = 3 biological replicates). In (**b**, **d**, **e**, **f**), different letters above the bar graphs indicate significant differences at $P < 0.05$ (One-way ANOVA). Source data are provided as a Source Data file.



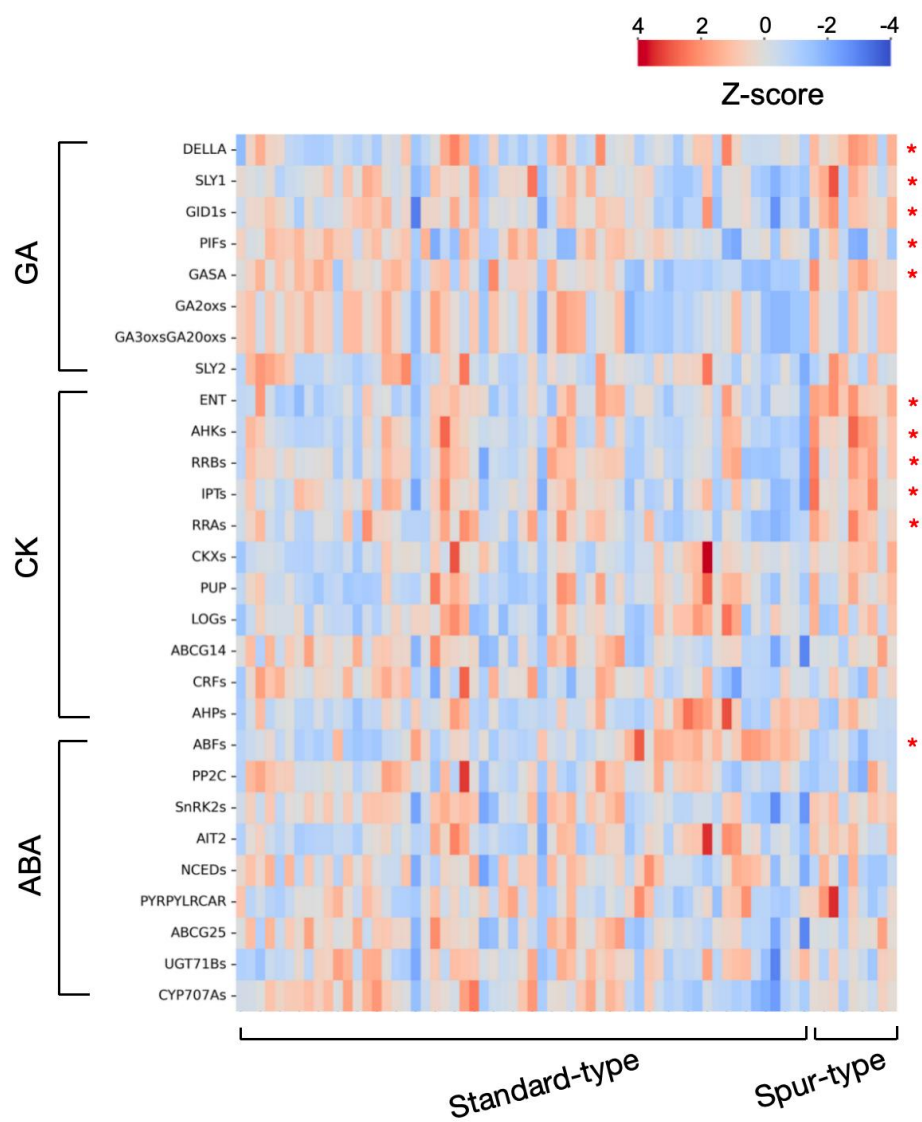
Supplementary Figure 19. Read mapping quality using different genome references. a GDDH13 genome used as the reference. **b** Fuji_Ral used as the reference. **c** Fuji_Del used as the reference. **d** Combined Fuji_Ral + Fuji_Del used as the reference without secondary read assignments. **e,f** Reads assigned to their appropriate haplotypes, Fuji_Ral (**e**) and Fuji_Del (**f**), using the pipeline developed in this study.



Supplementary Figure 20. Alignment scores using different genome references for read mapping. **a** GDDH13 used as the reference. **b** Fuji_Ral used as the reference. **c** Fuji_Del used as the reference. **d** Combined Fuji_Ral + Fuji_Del as the reference without secondary read assignments. **e,f** Reads assigned to their appropriate haplotypes, Fuji_Ral (**e**) and Fuji_Del (**f**), using the pipeline developed in this study.



Supplementary Figure 21. Number of mismatches using different genome references for read mapping. **a** GDDH13 used as the reference. **b** Fuji_Ral used as the reference. **c** Fuji_Del used as the reference. **d** Combined Fuji_Ral + Fuji_Del as the reference without secondary read assignments. **e,f** Reads assigned to their appropriate haplotypes, Fuji_Ral (**e**) and Fuji_Del (**f**), using the pipeline developed in this study.



Supplementary Figure 22. Expression patterns of genes related to GA, CK, and ABA hormones based on RNA-Seq data. Red asterisks indicate genes that show significant differences between standard-type and spur-type varieties ($P < 0.05$; Student's t -test). Source data are provided as a Source Data file.

Supplementary references

1. Li, W. *et al.* Near-gapless and haplotype-resolved apple genomes provide insights into the genetic basis of rootstock-induced dwarfing. *Nat Genet* **56**, 505–516 (2024).
2. Peng, H. *et al.* A haplotype-resolved genome assembly of *Malus domestica* ‘Red Fuji’. *Sci Data* **11**, 592 (2024).